

Chapter 19: Conic Sections (Ba Đường Conic)

Chapter Overview

Conic sections are curves obtained by intersecting a plane with a double cone. These elegant curves—**ellipse, hyperbola, and parabola**—have fascinated mathematicians since ancient Greece and appear throughout nature and technology.

From planetary orbits (Kepler's laws) to satellite dishes, architectural domes, and medical devices, conic sections are fundamental to understanding our world. This chapter explores their geometric definitions, standard equations, and real-world applications.

Real-world connections:

- **Astronomy:** Planetary orbits are ellipses with the Sun at one focus
- **Engineering:** Parabolic reflectors in telescopes and satellite dishes
- **Architecture:** Elliptical domes and hyperbolic cooling towers
- **Medicine:** Elliptical mirrors in lithotripsy (kidney stone treatment)
- **Physics:** Projectile motion follows parabolic paths

19.1 Bilingual Vocabulary (Từ vựng Song ngữ)

Core Concepts

English	IPA	Vietnamese	Example
Conic section	/ˈkɒnɪk ˈsekʃən/	Đường conic	The three main conic sections are ellipse, hyperbola, and parabola.
Ellipse	/ɪˈlɪps/	Elip	Planetary orbits are ellipses with the Sun at one focus.
Hyperbola	/haɪˈpɜːrbələ/	Hypebol	The cooling tower has a hyperbolic shape.

Parabola	/pəˈræbələ/	Parabol	The water fountain creates a parabolic arc.
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Key Elements

English	IPA	Vietnamese	Example
Focus (plural: foci)	/'fəʊkəs/ (/ 'fəʊsaɪ/)	Tiêu điểm	An ellipse has two foci.
Focal distance	/'fəʊkəl ˈdɪstəns/	Tiêu cự	The focal distance is $2c$.
Directrix	/dɪˈrektrɪks/	Đường chuẩn	A parabola is equidistant from its focus and directrix.
Focal parameter	/'fəʊkəl pəˈræmɪtər/	Tham số tiêu	The focal parameter p determines the parabola's width.

Equation Forms

English	IPA	Vietnamese	Example
Standard form	/'stændərd fɔːrm/	Phương trình chính tắc	The standard form of an ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
Semi-major axis	/'semi ˈmeɪdʒər ˈæksɪs/	Bán trục lớn	The semi-major axis is denoted by a .

Semi-minor axis	<i>/ˈsemi ˈmaɪnər ˈæksɪs/</i>	Bán trục nhỏ	<i>The semi-minor axis is denoted by b.</i>
Eccentricity	<i>/ˌɛksənˈtɹɪsəti/</i>	Độ lệch tâm	<i>The eccentricity measures how "stretched" the conic is.</i>

Properties

English	IPA	Vietnamese	Example
Vertex	<i>/ˈvɜːrteks/</i>	Đỉnh	<i>The parabola has one vertex at the origin.</i>
Center	<i>/ˈsentər/</i>	Tâm	<i>The center of the ellipse is at the origin.</i>
Asymptote	<i>/ˈæsɪmptoʊt/</i>	Tiệm cận	<i>A hyperbola has two asymptotes.</i>
Axis of symmetry	<i>/ˈæksɪs əv ˈsɪmətri/</i>	Trục đối xứng	<i>The parabola is symmetric about its axis.</i>

19.2 The Ellipse (Elip)

Definition

Ellipse Definition

An **ellipse** is the set of all points M in a plane such that the sum of distances from M to two fixed points

F_1

and

F_2

is constant:

$$MF_1 + MF_2 = 2a$$

where

$$a > 0$$

is a constant,

F_1

and

F_2

are the **foci** (singular: focus), and

$$F_1 F_2 = 2c$$

is the **focal distance** (tiêu cự).

Condition:

$$a > c$$

(the constant sum must be greater than the distance between foci).

Intuitive understanding: Imagine pinning two ends of a string at points

F_1

and

F_2

, then stretching the string with a pencil and moving it around—the pencil traces an ellipse.

Standard Form of Ellipse

Standard Equation of Ellipse

In a coordinate system with origin

O

at the midpoint of

$F_1 F_2$

and the

x

-axis along

$$F_1 F_2$$

, the ellipse has the standard form:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where

$$a > b > 0$$

, and:

- a
= semi-major axis (bán trục lớn)
- b
= semi-minor axis (bán trục nhỏ)
- $c = \sqrt{a^2 - b^2}$
(focal distance parameter)
- Foci:
 $F_1(-c, 0)$
and
 $F_2(c, 0)$
- Focal distance:
 $2c$

Key relationships:

- $a^2 = b^2 + c^2$
(Pythagorean-like relation)
- For any point
 M
on the ellipse:
 $MF_1 + MF_2 = 2a$

Properties of Ellipse

1. Vertices:

$$(\pm a, 0)$$

and

$$(0, \pm b)$$

2. **Center:**

$$(0, 0)$$

3. **Axes of symmetry:** Both

x

-axis and

y

-axis

4. **Domain:**

$$-a \leq x \leq a$$

5. **Range:**

$$-b \leq y \leq b$$

19.3 The Hyperbola (Hypebol)

Definition

Hyperbola Definition

A **hyperbola** is the set of all points

M

in a plane such that the absolute value of the difference of distances from

M

to two fixed points

F_1

and

F_2

is constant:

$$|MF_1 - MF_2| = 2a$$

where

$$a > 0$$

is a constant,

$$F_1$$

and

$$F_2$$

are the **foci**, and

$$F_1 F_2 = 2c$$

is the **focal distance**.

Condition:

$$a < c$$

(the constant difference must be less than the distance between foci).

Note: A hyperbola has **two branches**:

- One branch:

$$MF_1 - MF_2 = 2a$$

- Other branch:

$$MF_2 - MF_1 = 2a$$

Standard Form of Hyperbola

Standard Equation of Hyperbola

In a coordinate system with origin

O

at the midpoint of

$$F_1 F_2$$

and the

x

-axis along

$$F_1 F_2$$

, the hyperbola has the standard form:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where

$$a, b > 0$$

, and:

- $c = \sqrt{a^2 + b^2}$
(focal distance parameter)
- Foci:
 $F_1(-c, 0)$
and
 $F_2(c, 0)$
- Focal distance:
 $2c$
- For any point
 M
on the hyperbola:
 $|MF_1 - MF_2| = 2a$

Key relationships:

- $c^2 = a^2 + b^2$
- Asymptotes:
 $y = \pm \frac{b}{a}x$

Properties of Hyperbola

1. **Vertices:**
 $(\pm a, 0)$
2. **Center:**
 $(0, 0)$
3. **Asymptotes:**
 $y = \pm \frac{b}{a}x$
(the branches approach these lines)
4. **Axes of symmetry:** Both
 x
-axis and

y
-axis

5. **Domain:**

$$x \leq -a$$

or

$$x \geq a$$

19.4 The Parabola (Parabol)

Definition

Parabola Definition

A **parabola** is the set of all points

M

in a plane that are equidistant from a fixed point

F

(the **focus**) and a fixed line

Δ

(the **directrix**) that does not pass through

F

:

$$MF = d(M, \Delta)$$

The distance from

F

to

Δ

is called the **focal parameter**

p

.

Standard Form of Parabola

Standard Equation of Parabola

In a coordinate system with origin

O

at the midpoint of the perpendicular from

F

to

Δ

, and the

x

-axis along this perpendicular (pointing from

Δ

to

F

), the parabola has the standard form:

$$y^2 = 2px$$

where

$$p > 0$$

is the focal parameter, and:

- Focus:

$$F\left(\frac{p}{2}, 0\right)$$

- Directrix:

$$x = -\frac{p}{2}$$

- Vertex:

$$(0, 0)$$

Note: The parabola

$$y = ax^2$$

can be written in standard form by rotation and translation.

Properties of Parabola

1. **Vertex:**

$$(0, 0)$$

2. **Axis of symmetry:**

x
-axis

3. **Focus:**

$$\left(\frac{p}{2}, 0\right)$$

4. **Directrix:**

$$x = -\frac{p}{2}$$

5. **Domain:**

$$x \geq 0$$

6. **Range:** All real numbers

19.5 Summary Table of Conic Sections

Conic	Standard Form	Foci	Key Relation	Condition
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$(\pm c, 0)$	$MF_1 + MF_2 = 2a$	$a > b > 0$, $c = \sqrt{a^2 - b^2}$
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$(\pm c, 0)$	$ MF_1 - MF_2 = 2a$	$a, b > 0$, $c = \sqrt{a^2 + b^2}$
Parabola	$y^2 = 2px$	$\left(\frac{p}{2}, 0\right)$	$MF = d(M, \Delta)$	$p > 0$

Memory aid:

- **Ellipse:** Sum ($+$) in definition $\rightarrow$ Plus (

- **Ellipse:** Sum of distances → One variable squared
 $+$
) in equation
 - **Hyperbola:** Difference ()
 $-$
) in definition → Minus ()
 $-$
) in equation
 - **Parabola:** Equal distances → One variable squared
-

19.6 Worked Examples

Example 1: Finding Elements of an Ellipse

Problem: Given the ellipse

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

, find:

- The foci and focal distance
- The sum of distances from any point on the ellipse to the two foci

Solution:

- From the equation, we have:

- $a^2 = 25 \Rightarrow a = 5$
- $b^2 = 16 \Rightarrow b = 4$
- $c = \sqrt{a^2 - b^2} = \sqrt{25 - 16} = \sqrt{9} = 3$

Therefore:

- Foci:
 $F_1(-3, 0)$
 and
 $F_2(3, 0)$
- Focal distance:
 $2c = 6$

b) For any point

M

on the ellipse:

$$MF_1 + MF_2 = 2a = 2(5) = 10$$

Answer: Foci at

$(\pm 3, 0)$

, focal distance = 6, sum of distances = 10.

Example 2: Finding the Equation of an Ellipse

Problem: Find the standard equation of an ellipse with foci at

$F_1(-4, 0)$

and

$F_2(4, 0)$

, and the sum of distances from any point on the ellipse to the foci is 10.

Solution:

From the given information:

- Focal distance:

$$2c = 8 \Rightarrow c = 4$$

- Sum of distances:

$$2a = 10 \Rightarrow a = 5$$

Using the relation

$$c^2 = a^2 - b^2$$

:

$$b^2 = a^2 - c^2 = 25 - 16 = 9$$

Therefore, the equation is:

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

Answer:

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

Example 3: Finding Elements of a Hyperbola

Problem: Given the hyperbola

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

, find:

- a) The foci and focal distance
- b) The absolute value of the difference of distances from any point on the hyperbola to the two foci
- c) The equations of the asymptotes

Solution:

a) From the equation:

- $a^2 = 9 \Rightarrow a = 3$
- $b^2 = 16 \Rightarrow b = 4$
- $c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

Therefore:

- Foci:
 $F_1(-5, 0)$
and
 $F_2(5, 0)$
- Focal distance:
 $2c = 10$

b) For any point

M

on the hyperbola:

$$|MF_1 - MF_2| = 2a = 2(3) = 6$$

c) Asymptotes:

$$y = \pm \frac{b}{a}x = \pm \frac{4}{3}x$$

Answer: Foci at

$$(\pm 5, 0)$$

, focal distance = 10, difference = 6, asymptotes:

$$y = \pm \frac{4}{3}x$$

Example 4: Finding the Equation of a Hyperbola

Problem: Find the standard equation of a hyperbola with foci at

$$F_1(-5, 0)$$

and

$$F_2(5, 0)$$

, and the absolute value of the difference of distances from any point on the hyperbola to the foci is 6.

Solution:

From the given information:

- Focal distance:

$$2c = 10 \Rightarrow c = 5$$

- Difference:

$$2a = 6 \Rightarrow a = 3$$

Using the relation

$$c^2 = a^2 + b^2$$

:

$$b^2 = c^2 - a^2 = 25 - 9 = 16$$

Therefore, the equation is:

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

Answer:

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

Example 5: Finding Elements of a Parabola

Problem: Given the parabola

$$y^2 = 8x$$

, find:

- a) The focus and directrix
- b) The points on the parabola whose distance to the focus is 5

Solution:

a) From

$$y^2 = 8x$$

, we have

$$2p = 8 \Rightarrow p = 4$$

Therefore:

- Focus:

$$F\left(\frac{p}{2}, 0\right) = F(2, 0)$$

- Directrix:

$$x = -\frac{p}{2} = -2$$

b) Let

$$M(x_0, y_0)$$

be a point on the parabola with

$$MF = 5$$

.

Since

$$M$$

is on the parabola:

$$MF = d(M, \Delta)$$

The directrix is

$$x = -2$$

, so:

$$d(M, \Delta) = x_0 - (-2) = x_0 + 2$$

Therefore:

$$x_0 + 2 = 5 \Rightarrow x_0 = 3$$

From

$$y_0^2 = 8x_0 = 8(3) = 24$$

:

$$y_0 = \pm\sqrt{24} = \pm 2\sqrt{6}$$

Answer: Focus at

$$(2, 0)$$

, directrix

$$x = -2$$

; points:

$$(3, 2\sqrt{6})$$

and

$$(3, -2\sqrt{6})$$

Example 6: Identifying Conic Sections

Problem: Identify each conic section and find its key elements:

a)

$$\frac{x^2}{16} + \frac{y^2}{25} = 1$$

b)

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

c)

$$x^2 = 12y$$

Solution:

a) This is an **ellipse** (both terms positive, different denominators).

- Since

$$25 > 16$$

, we have

$$a^2 = 25$$

,

$$b^2 = 16$$

- $a = 5$

,

$$b = 4$$

,

$$c = \sqrt{25 - 16} = 3$$

- But the larger denominator is under

$$y^2$$

, so the major axis is vertical

- Foci:

$$(0, \pm 3)$$

b) This is a **hyperbola** (one positive, one negative term).

- $a^2 = 4$

,

$$b^2 = 9$$

- $a = 2$

,

$$b = 3$$

,

$$c = \sqrt{4 + 9} = \sqrt{13}$$

- Foci:

$$(\pm \sqrt{13}, 0)$$

- Asymptotes:

$$y = \pm \frac{3}{2}x$$

c) Rewrite as

$$x^2 = 12y$$

(one variable squared).

This is a **parabola** opening upward.

- $2p = 12 \Rightarrow p = 6$

- Focus:
 $(0, 3)$

- Directrix:
 $y = -3$

Answer: (a) Ellipse with vertical major axis, foci at $(0, \pm 3)$
 ; (b) Hyperbola, foci at $(\pm \sqrt{13}, 0)$
 ; (c) Parabola, focus at $(0, 3)$

Example 7: Ellipse Through Given Points

Problem: Find the standard equation of an ellipse passing through points

$A(2, 0)$

and

$B(0, 3)$

.

Solution:

The standard form is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

.

Since

$A(2, 0)$

is on the ellipse:

$$\frac{4}{a^2} + \frac{0}{b^2} = 1 \Rightarrow a^2 = 4$$

Since

$B(0, 3)$

is on the ellipse:

$$\frac{0}{a^2} + \frac{9}{b^2} = 1 \Rightarrow b^2 = 9$$

Therefore, the equation is:

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

Answer:

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

Example 8: Hyperbola with Given Asymptotes

Problem: Find the standard equation of a hyperbola with asymptotes

$$y = \pm 2x$$

and passing through point

$$P(2, 3)$$

.

Solution:

For a hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

, the asymptotes are

$$y = \pm \frac{b}{a}x$$

.

Given asymptotes

$$y = \pm 2x$$

:

$$\frac{b}{a} = 2 \Rightarrow b = 2a$$

Since

$$P(2, 3)$$

is on the hyperbola:

$$\frac{4}{a^2} - \frac{9}{b^2} = 1$$

Substituting

$$b = 2a$$

:

$$\frac{4}{a^2} - \frac{9}{4a^2} = 1$$

$$\frac{16-9}{4a^2} = 1$$

$$\frac{7}{4a^2} = 1 \Rightarrow a^2 = \frac{7}{4}$$

$$b^2 = 4a^2 = 7$$

Therefore, the equation is:

$$\frac{x^2}{\frac{7}{4}} - \frac{y^2}{7} = 1$$

or equivalently:

$$\frac{4x^2}{7} - \frac{y^2}{7} = 1$$

Answer:

$$\frac{4x^2}{7} - \frac{y^2}{7} = 1$$

or

$$4x^2 - y^2 = 7$$

19.7 Applications of Conic Sections

Application 1: Planetary Orbits (Kepler's First Law)

Problem: The Earth's orbit around the Sun is an ellipse with the Sun at one focus. The semi-major axis is approximately

$$a = 149.6$$

million km, and the semi-minor axis is approximately

$$b = 149.1$$

million km. Find:

a) The distance

c

from the Sun to the center of the ellipse

b) The closest and farthest distances from Earth to the Sun

Solution:

a) Using

$$c = \sqrt{a^2 - b^2}$$

:

$$c = \sqrt{149.6^2 - 149.1^2} = \sqrt{22380.16 - 22230.81} = \sqrt{149.35} \approx 12.2 \text{ million km}$$

b) The closest distance (perihelion) is:

$$a - c = 149.6 - 12.2 = 137.4 \text{ million km}$$

The farthest distance (aphelion) is:

$$a + c = 149.6 + 12.2 = 161.8 \text{ million km}$$

Answer:

$$c \approx 12.2$$

million km; closest distance ≈ 137.4 million km, farthest ≈ 161.8 million km

Application 2: Parabolic Satellite Dish

Problem: A satellite dish has a parabolic cross-section. The receiver is placed at the focus, which is 30 cm from the vertex. If the dish is 120 cm wide at the opening, how deep is the dish?

Solution:

Place the vertex at the origin with the parabola opening to the right:

$$y^2 = 2px$$

.

The focus is at

$$\left(\frac{p}{2}, 0\right) = (30, 0)$$

, so:

$$\frac{p}{2} = 30 \Rightarrow p = 60$$

The equation is:

$$y^2 = 120x$$

At the opening, the width is 120 cm, so

$$y = \pm 60$$

cm.

Substituting

$$y = 60$$

:

$$3600 = 120x \Rightarrow x = 30$$

The depth of the dish is 30 cm.

Answer: The dish is 30 cm deep.

Application 3: Hyperbolic Navigation (LORAN)

Problem: Two radio stations

A

and

B

are located 300 km apart. A ship receives signals from both stations. The signal from

A

arrives 0.0005 seconds before the signal from

B

. If radio waves travel at 300,000 km/s, find the equation of the hyperbola on which the ship is located (with

A

and

B

as foci).

Solution:

The time difference is

$$\Delta t = 0.0005$$

s.

The distance difference is:

$$|SA - SB| = v \cdot \Delta t = 300000 \times 0.0005 = 150 \text{ km}$$

So

$$2a = 150 \Rightarrow a = 75$$

km.

The distance between stations is

$$2c = 300 \Rightarrow c = 150$$

km.

Using

$$b^2 = c^2 - a^2$$

:

$$b^2 = 150^2 - 75^2 = 22500 - 5625 = 16875$$

Placing

A

and

B

at

$(\pm 150, 0)$

, the equation is:

$$\frac{x^2}{5625} - \frac{y^2}{16875} = 1$$

Answer:

$$\frac{x^2}{5625} - \frac{y^2}{16875} = 1$$

19.8 Optical Properties of Conics

Conic sections have remarkable **optical (or reflective) properties** that make them invaluable in technology:

Ellipse Reflective Property

Ellipse Reflection

A light ray emanating from one focus of an ellipse will reflect off the ellipse and pass through the other focus.

Applications:

- **Lithotripsy:** Shock waves from one focus converge at the other focus to break kidney stones
- **Whispering galleries:** Sound from one focus can be heard clearly at the other focus

Parabola Reflective Property

Parabola Reflection

A light ray parallel to the axis of a parabola will reflect off the parabola and pass through the focus. Conversely, a ray from the focus reflects parallel to the axis.

Applications:

- **Satellite dishes:** Incoming parallel signals converge at the focus (receiver)
- **Telescopes:** Parabolic mirrors focus parallel light rays from distant stars
- **Flashlights:** Light from a bulb at the focus reflects as parallel beams

Hyperbola Reflective Property

Hyperbola Reflection

A light ray directed toward one focus of a hyperbola will reflect off the hyperbola as if coming from the other focus.

Applications:

- **Telescope design:** Cassegrain telescopes use hyperbolic secondary mirrors
- **Cooling towers:** Hyperbolic shape provides structural strength

19.9 Practice Exercises

Exercise 1: Find the foci, focal distance, and vertices of the ellipse:

$$\frac{x^2}{36} + \frac{y^2}{20} = 1$$

Exercise 2: Find the standard equation of an ellipse with foci at

$$(\pm 3, 0)$$

and vertices at

$$(\pm 5, 0)$$

.

Exercise 3: Find the foci, vertices, and asymptotes of the hyperbola:

$$\frac{y^2}{25} - \frac{x^2}{9} = 1$$

Exercise 4: Find the focus and directrix of the parabola

$$y^2 = 16x$$

Exercise 5: A parabolic arch has a span of 20 meters and a maximum height of 10 meters. Find the equation of the parabola (with vertex at the top).

19.10 Chapter Summary

Key Concepts

1. **Ellipse:** Sum of distances to two foci is constant

- Standard form:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(

$$a > b > 0$$

)

- Relation:

$$c^2 = a^2 - b^2$$

- Foci:

$$(\pm c, 0)$$

2. **Hyperbola:** Absolute difference of distances to two foci is constant

- Standard form:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

- Relation:

$$c^2 = a^2 + b^2$$

- Foci:

$$(\pm c, 0)$$

- Asymptotes:

$$y = \pm \frac{b}{a}x$$

3. **Parabola:** Distance to focus equals distance to directrix

- Standard form:

$$y^2 = 2px$$

(

$$p > 0$$

)

- Focus:

$$\left(\frac{p}{2}, 0\right)$$

- Directrix:

$$x = -\frac{p}{2}$$

Quick Comparison

Property	Ellipse	Hyperbola	Parabola
Foci	2	2	1
Equation sign	+	–	One variable
c relation	$c^2 = a^2 - b^2$	$c^2 = a^2 + b^2$	N/A
Asymptotes	None	$y = \pm \frac{b}{a}x$	None
Closed curve	Yes	No (two branches)	No (one branch)

19.11 Review Exercises

Exercise 1: Identify the conic section and find its key elements:

a)

$$\frac{x^2}{49} + \frac{y^2}{25} = 1$$

b)

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

c)

$$x^2 = -8y$$

Exercise 2: Find the standard equation of the conic:

a) Ellipse with foci at

$$(0, \pm 4)$$

and vertices at

$$(0, \pm 5)$$

b) Hyperbola with vertices at

$$(\pm 3, 0)$$

and foci at

$$(\pm 5, 0)$$

c) Parabola with focus at

$$(0, 2)$$

and directrix

$$y = -2$$

Exercise 3: An elliptical pool table has foci 2 meters apart. If a ball is placed at one focus and struck, it will always pass through the other focus after bouncing off the edge. If the table is 3 meters long (major axis), find the equation of the ellipse.

Exercise 4: A cooling tower has a hyperbolic shape. At its narrowest point (the "neck"), the radius is 20 meters. At a height 30 meters above the neck, the radius is 30 meters. If the center of the hyperbola is at the neck, find the equation of the hyperbola.

Exercise 5: A parabolic reflector has a diameter of 80 cm and a depth of 20 cm. Where should the light source be placed to produce a parallel beam?

19.12 Memory Aids

Remembering the Formulas

Ellipse vs. Hyperbola:

- **Ellipse:** "Plus" in equation ($+$), "c is less" ($c < a$),

$$c^2 = a^2 - b^2$$

- **Hyperbola:** "Minus" in equation (

—

), "c is more" (

$$c > a$$

),

$$c^2 = a^2 + b^2$$

Parabola:

- "

$$y^2 = 2px$$

" → Focus at

$$\left(\frac{p}{2}, 0\right)$$

, directrix at

$$x = -\frac{p}{2}$$

- "Half the coefficient of
 x
 " gives the focus distance

Identifying Conics from Equations

1. Both

$$x^2$$

and

$$y^2$$

with same sign → Ellipse (or circle if coefficients equal)

2. Both

$$x^2$$

and

$$y^2$$

with opposite signs → Hyperbola

3. Only one variable squared → Parabola

Real-World Associations

- **Ellipse:** Planets, orbits, whispering galleries
- **Hyperbola:** Cooling towers, navigation (LORAN), telescope mirrors

- **Parabola:** Satellite dishes, projectile motion, flashlight reflectors

19.13 Common Mistakes to Avoid

⚠ **Mistake 1:** Confusing

$$c^2 = a^2 - b^2$$

(ellipse) with

$$c^2 = a^2 + b^2$$

(hyperbola)

- **Remember:** Ellipse uses subtraction (

–

), hyperbola uses addition (

+

)

⚠ **Mistake 2:** Forgetting the condition

$$a > b$$

for ellipse standard form

- If

$$b > a$$

, the major axis is vertical, not horizontal

⚠ **Mistake 3:** Writing parabola focus as

$$(p, 0)$$

instead of

$$\left(\frac{p}{2}, 0\right)$$

- The focus is at **half** the focal parameter distance

⚠ **Mistake 4:** Confusing "sum" (ellipse) with "difference" (hyperbola) in definitions

- Ellipse:

$$MF_1 + MF_2 = 2a$$

- Hyperbola:

$$|MF_1 - MF_2| = 2a$$

⚠ **Mistake 5:** Forgetting absolute value in hyperbola definition

- The difference can be positive or negative, so use $|MF_1 - MF_2|$
-

 **Congratulations!** You have completed Chapter 19: Conic Sections. You now understand the three classical conic sections—ellipse, hyperbola, and parabola—their equations, properties, and applications in science, engineering, and everyday life.